

Lab tutorial for Chapter 1: Combinatorial Analysis

Exercise 1.

- (a) if a 6—die is rolled and a coin is flipped at the same time, how many outcomes are possible?
- (b) If five 6— dice are rolled, how many outcomes are possible?

Solution 1.

- (a) Since a die has 6 outcomes and a coin has 2 outcomes, hence there is a total of $6 \times 2 = 12$ outcomes if they are performed together.
- (b) Since a die has 6 outcomes and it tossed 5 times, there a total of $6^5 = 7776$

Exercise 2.

- (a) Determine the number of arrangements of the letters in the word DUPLICATES
- (b) Determine the number of arrangements of the letters in the word REARRANGE

Solution 2.

- (a) Since there is no duplicated letters in the 10—letters word DUPLICATES, there are $10! = 3628800$ different arrangements of the letters.
- (b) Since there are 9 letters in the word REARRANGE, with the R repeated 3 times, and the E and A both repeated twice, there are

$$\frac{9!}{3!2!2!} = 15120$$

Exercise 3.

- (a) If there are 20 patients in a given ward, and each nurse is assigned to a rotation of 5 patients, how many different rotation assignments are possible for the first nurse?
- (b) In a race of 8 runners, how many different medal results (i.e., top-3 finishes: gold, silver, bronze) are possible

Solution 3.

- (a) We are choosing 5 patients from a group of 20, without regard to order. This is calculated using the number of combinations:

$$C_5^{20} = \frac{20!}{5!(20-5)!} + \frac{20 \times 19 \times 18 \times 17 \times 16}{5 \times 4 \times 3 \times 2 \times 1} = \frac{1860480}{120} = 15504$$

So, there are **15,504 different rotation assignments** possible for the first nurse.

- (b) There 8 option for the first place, 7 for the second place and 6 for the third place, hence there are a total of

$$8 \times 7 \times 6 \quad \text{or} \quad P_3^8 = \frac{8!}{5!} = 336$$

unique medal results.

Exercise 4.

- (a) From a standard deck of 52 cards, how many different 5-card poker “hands” are possible?
- (b) In LOTTO-649, 6 winning numbers are randomly selected from 1 – 49 how many different winning combinations are possible?
- (c) If 12 students apply to be on a 4- person yearbook committee, how many outcomes are possible?

Solution 4.

(a) Since order is not relevant, there are

$$C_5^{52} = \frac{52!}{5!(52-5)!} = 2598960$$

5-cards poker hands.

(b) Since order is not relevant, there are

$$C_6^{49} = \frac{49!}{6!(49-6)!} = 13983816$$

unique possible winning combinations.

(c) For the committee membership the order in which the students are selected is irrelevant, so there are

$$C_4^{12} = \frac{12!}{4!(12-4)!} = 495$$

different possible committee membership outcomes.

Exercise 5.

How many subsets are there of a set consisting of n elements?

Solution 5.

Since there are $\binom{n}{k}$ subsets of size k , the desired answer is

$$\sum_{k=0}^n \binom{n}{k} = (1+1)^n = 2^n.$$

Exercise 6.

Using binomial identity, show that

(a)

$$\sum_{r=0}^n (-1)^r \binom{n}{r} = 0.$$

(b)

$$\sum_{r=0}^n \binom{n}{r} (a-1)^r = a^n.$$

Solution 6.

Recall that the binomial identity is given by

$$(x+y)^n = \sum_{r=0}^n \binom{n}{r} x^r y^{n-r}. \quad (1)$$

(a) If we set $x = -1, y = 1$, then (1) leads to

$$\sum_{r=0}^n (-1)^r \binom{n}{r} = \sum_{r=0}^n \binom{n}{r} (-1)^r 1^{n-r} = (-1+1)^n = 0.$$

(b) Now setting $x = a-1, y = 1$, then (1) leads to

$$\sum_{r=0}^n \binom{n}{r} (a-1)^r = \sum_{r=0}^n \binom{n}{r} (a-1)^r 1^{n-r} = (a-1+1)^n = a^n.$$

Exercise 7.

Applying $\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$ repeatedly, show that

$$\binom{n}{r} = \sum_{i=1}^{r+1} \binom{n-i}{r-i+1}.$$

Solution 7.

If we write down the right side explicitly,

$$\sum_{i=1}^{r+1} \binom{n-i}{r-i+1} = \binom{n-1}{r} + \binom{n-2}{r-1} + \cdots + \binom{n-r-1}{0}.$$

Therefore, we have

$$\begin{aligned}
\binom{n}{r} &= \binom{n-1}{r} + \binom{n-1}{r-1} \\
&= \binom{n-1}{r} + \binom{n-2}{r-1} + \binom{n-2}{r-2} \\
&= \binom{n-1}{r} + \binom{n-2}{r-1} + \binom{n-3}{r-2} + \binom{n-3}{r-3} \\
&= \dots \\
&= \binom{n-1}{r} + \binom{n-2}{r-1} + \binom{n-3}{r-2} + \dots + \binom{n-r}{1} + \binom{n-r}{0} \\
&= \binom{n-1}{r} + \binom{n-2}{r-1} + \binom{n-3}{r-2} + \dots + \binom{n-r}{1} + \binom{n-r-1}{0} \\
&= \sum_{i=1}^{r+1} \binom{n-i}{r-i+1}.
\end{aligned}$$

Exercise 8.

The price of a European tour included four stopovers to be selected from among 10 cities. In how many different ways can one plan such a tour

- (a) if the order of stopovers matters;
- (b) if the order of stopovers does not matter?

Solution 8.

- (a) This is corresponding to number of permutations of 10 distinct elements taking them 4 at a time,

$$P_4^{10} = \frac{10!}{6!} = 10 \times 9 \times 8 \times 7 = 5040.$$

- (b) This is corresponding to number of combinations of 10 distinct elements taking them 4 at a time,

$$C_4^{10} = \frac{10!}{4!6!} = \frac{5040}{4!} = 210.$$

Exercise 9.

A shipment of 10 television sets includes three that are defective. In how many ways can a hotel purchase four of these sets and receive at least two of the defective sets?

Solution 9.

Consider two cases, (1) there are exactly 2 defective sets, and 2 working sets; (2) there are 3 defective sets, and 1 working set. Therefore, under (1), we have

$$\binom{3}{2} \binom{7}{2} = 3 \times 21 = 63.$$

Under (2), we have

$$\binom{3}{3} \binom{7}{1} = 1 \times 7 = 7.$$

Finally, we have

$$63 + 7 = 70$$

possible ways.